

PERCENTAGES AND QUANTITIES

Percentages are a useful way of expressing a value as a proportion of the total number. If two out of three of a percentage, value, and total number are known, it is possible to find out the missing quantity using arithmetic.

Finding an amount as a % of another amount

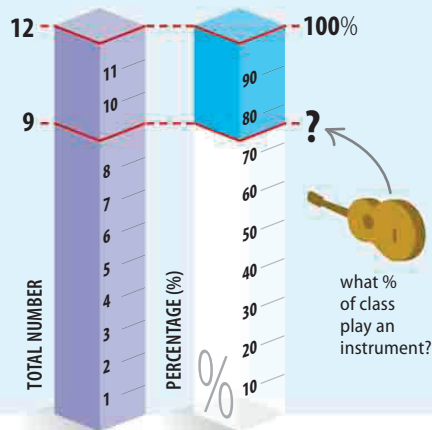
Out of 12 pupils in a class, 9 play a musical instrument. To find the known value (9) as a percentage of the total (12), divide the known value by the total number and multiply by 100.

$$\frac{\text{Number to turn into a \%}}{\text{Total number}} \times 100 = \% \text{ of total number}$$

$$\frac{9}{12} \times 100 = 75\% \text{ play instruments}$$

Divide the known number by the total number ($9 \div 12 = 0.75$).

Multiply the result by 100 to get the percentage ($0.75 \times 100 = 75$).



Finding the total number from a %

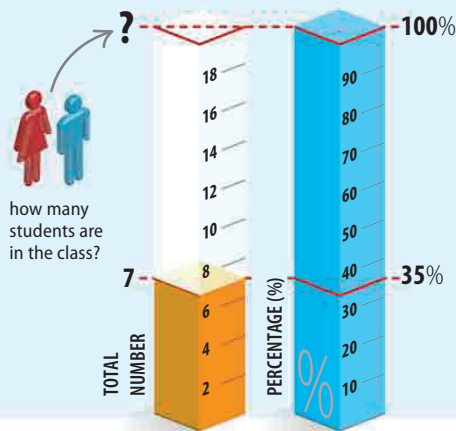
In a class, 7 children make up 35% of the total. To find the total number of students in the class, divide the known value (7) by the known percentage (35) and multiply by 100.

$$\frac{\text{Known amount}}{\text{Known \%}} \times 100 = \text{Total number}$$

$$\frac{7}{35} \times 100 = 20 \text{ students}$$

Divide the known amount by the known percentage ($7 \div 35 = 0.2$).

Multiply the result by 100 to get the total number ($0.2 \times 100 = 20$).



REAL WORLD

Percentages

Percentages are all around us—in stores, in newspapers, on TV—everywhere. Many things in everyday life are measured and compared in percentages—how much an item is reduced in a sale; what the interest rate is on a mortgage or a bank loan; or how efficient a light bulb is by the percentage of electricity it converts to light. Percentages are even used to show how much of the recommended daily intake of vitamins and other nutrients is in food products.

